Linear Algebra & Calculus for Deep Learning

Advanced Machine Learning

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DRAFT

This slide deck is still under active development. Expect changes before it is finalized.

Learning objectives

- Build a concrete, geometric sense of what a "dimension" is — a feature vector vs. a multi-dimensional measurement
- Compute partial derivatives and assemble them into a gradient
- See the chain rule compose two functions — and watch autograd automate exactly that
- Use estimators (mean, std) to standardize features, and know what "estimator" means

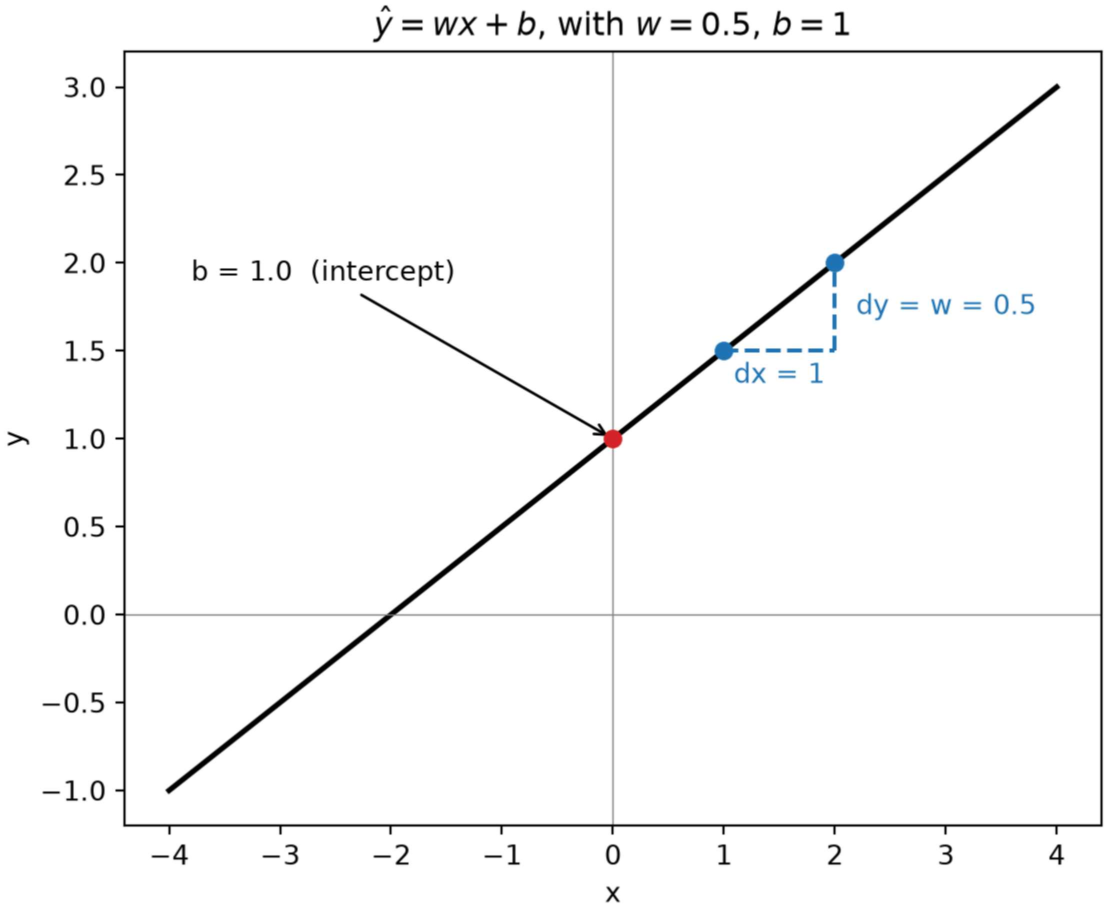
Optimizing by guessing

What is a model?

- A model is a function with **parameters** we can tune
- The simplest useful one is linear:

$$\hat{y} = wx + b$$

- `w` — slope: how much `ŷ` changes per unit of `x`
- `b` — intercept: where the line sits when `x = 0`
- This is the same `Ax + b` you've already seen — just one input, one output



Quick aside: this generalizes to vectors

- The example above used scalars — one `x`, one `w` — to keep things concrete
- In general, a model takes a whole feature vector **x** and a matching weight vector **w**:

$$\hat{y} = \mathbf{w} \cdot \mathbf{x} + b$$

- Same idea, same shape (`Ax + b`) — just more numbers doing the same job
- We'll build up exactly what **w** and **x** mean, and what that `·` is doing, in a few slides

How do we improve a model?

- We have a way to score any choice of `(w, b)`: mean squared error
- So… how do we pick a *better* `(w, b)`?

Simplest possible idea: nudge them randomly, and see what happens.

Random search, live

This works… eventually

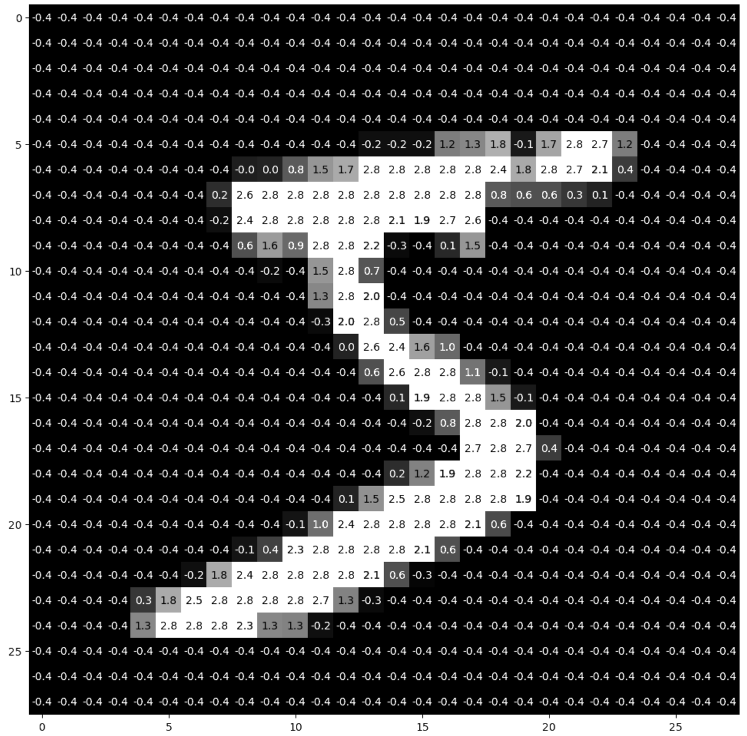
- Blue lines improved the fit, red lines made it worse — and by default, we take the step either way
- Check “only keep steps that improve” — about half the attempts still get thrown away (red), but it actually converges *faster*, because it never backslides
- Either way: there’s no sense of *direction*, only whether we got lucky
- With only two parameters this is already slow. Real models have millions.

The question of the week: what if `choose_next_value()` didn’t have to guess?

Vectors, matrices, dimensions

What does “dimension” mean?

- Dimension means how many distinct input values we have
  - One house: `[sqft, bedrooms, age]`



- It can also describe the shape of our data
  - A greyscale image that is 28x28 would have shape `(1, 28, 28)`, so 784 input dimensions
  - It is also “2D” in the colloquial sense

> **Note**
>
> We can have both of these combined! For instance, `[sqft, bedrooms, age, (1, 28, 28)]`

A feature vector

```python
import torch

house = torch.tensor([1800.0, 3, 12])  # sqft, bedrooms, age
print(house.shape)
```

```
torch.Size([3])
```

- 3 numbers, 3 *different* things — shape `(3,)`
- Position matters because we’ve now chosen a convention for the data
  - In this dataset:
    - index 0 always means sqft
    - index 1 always means bedrooms
    - index 2 always means age

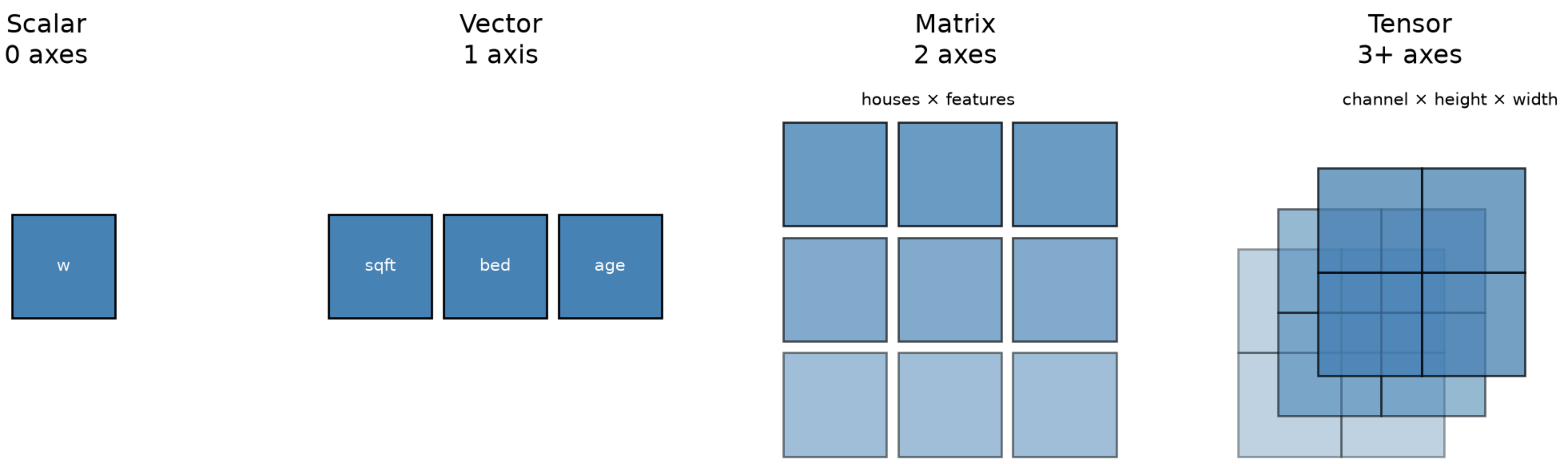
A multi-dimensional measurement

```python
image = torch.zeros(1, 28, 28)  # channels, height, width
print(image.shape)
```

```
torch.Size([1, 28, 28])
```

- Unlike the feature vector, no axis here is a fundamentally different *kind* of thing
  - Axis 0 is “which channel,” axis 1 is “which row,” axis 2 is “which column” — all just *more pixels*
- `.shape` tells you the size of each axis, but not which axis is which — that’s a convention you have to know separately (here: channel, then height, then width)

Tensors: one word for all of these



- **Scalar** (0 axes) — a single number, like our `w`
- **Vector** (1 axis) — our feature vector `[sqft, bedrooms, age]`
- **Matrix** (2 axes) — a *batch* of feature vectors, one row per house
- **Tensor** (3+ axes) — our image: channel, height, width

- “Tensor” isn’t new math — it’s just the general word for “however many axes you need”
- Once you think in tensors, you stop asking “is this a vector or a matrix?” and just ask “what does each axis mean?”

Important Tensor Math

How big is a vector? Norms

- **L2 norm** — straight-line length:

$$\|\mathbf{x}\|_2 = \sqrt{\sum_i x_i^2}$$

- **L1 norm** — total distance if you could only move along axes:

$$\|\mathbf{x}\|_1 = \sum_i |x_i|$$

```
x = torch.tensor([3.0, 4.0])
print(torch.linalg.norm(x, ord=2))
print(torch.linalg.norm(x, ord=1))
```

```
tensor(5.)
tensor(7.)
```

Does $\|\texttt{pred - actual}\|_2^2$ look familiar? That's exactly what mean squared error is measuring. More on this when we get to loss functions.

Why norms matter: measuring importance

- “Size” here rarely means literal length — it usually means **relative importance**
  - A large weight → that feature has a big influence on the prediction
  - A large gradient → that parameter is about to change a lot
  - A large error vector → the model is very wrong on this example
- We’ll spend real time controlling these sizes *on purpose*: regularization shrinks large weights, gradient clipping caps large gradients

Dot product: two ways to think about it

- **Algebraic:** sum of elementwise products

$$\mathbf{a} \cdot \mathbf{b} = \sum_i a_i b_i$$

- **Geometric:** a measure of *alignment* — large when two vectors point the same way, near zero when they're perpendicular, negative when they point opposite ways

```
a = torch.tensor([1.0, 0.0])
b = torch.tensor([1.0, 0.0])
c = torch.tensor([0.0, 1.0])

print(torch.dot(a, b))  # same direction
print(torch.dot(a, c))  # perpendicular
```

```
tensor(1.)
tensor(0.)
```

- This dual meaning — "sum of products" *and* "similarity" — comes back constantly: cosine similarity, attention, and beyond

Why dot products matter: measuring agreement

- In plain terms, a dot product asks **“how much do these two things agree?”**
  - Pointing the same way → strong agreement (big positive number)
  - Perpendicular → no relationship (near zero)
  - Pointing opposite ways → active disagreement (big negative number)
- This is exactly how a model *evaluates* an input: each output is a dot product between the input and a learned weight vector — “how well does this input match what I’m looking for?”

Matrix-vector product: a batch of dot products

- Multiplying a matrix by a vector is just *many dot products done at once*, one per row

```
A = torch.tensor([[1.0, 0.0],
                  [0.0, 1.0],
                  [1.0, 1.0]])
x = torch.tensor([2.0, 3.0])

print(A @ x)
print(torch.dot(A[2], x))  # row 2 of A, on its own
```

```
tensor([2., 3., 5.])
tensor(5.)
```

- Row `i` of `A @ x` is exactly `A[i] · x` — nothing new, just done for every row together
- This is what a linear layer computes: one dot product per output neuron, in parallel

What this means: a matrix is a function

- $\mathbf{Ax}$ isn't just "multiply a grid by a list" — it's **evaluating** $\mathbf{x}$ using $\mathbf{A}$
- Read it the same way you'd read $f(x)$: "put $\mathbf{x}$ into $\mathbf{A}$"
- Every row of $\mathbf{A}$ is a dot product "looking for" something — the matrix-vector product asks that "how much do you agree?" question for every row, all at once
- That's the entire forward pass of a linear layer, in one line:

$$\mathbf{Ax} + \mathbf{b}$$

Calculus: derivatives, partials, gradients

Derivatives: a fast recap

- The derivative of f at a point is its **slope** there — how fast f changes as x changes
  - Notation: $f'(x)$, or $\frac{df}{dx}$ – this latter one is much more explicit
  - $\frac{df}{dx}$ can be understood as “ratio of delta-f to delta-x” where “delta” is how we say “change in”

We only need a handlefule of rules:

Power Rule	$\frac{d}{dx}x^n = nx^{(n-1)}$
Constant Rule	$\frac{d}{dx}c = 0$
Constant Multiple Rule	$\frac{d}{dx}\big(c \cdot f(x)\big) = c \cdot f'(x)$
Chain Rule	$\frac{d}{dx}f(g(x)) = f'(g(x)) \cdot g'(x)$

The Chain Rule: Composing two functions

- $g(x) = 2x$
- $f(g) = g^2$
- Compose them: $f(g(x)) = (2x)^2 = 4x^2$

```
def g(x):
    return 2 * x


def f(g_val):
    return g_val ** 2


x = 3.0
print(f(g(x)))
```

```
36.0
```

Two ways to get the same derivative

- **Direct:** differentiate the composed function outright

$$\frac{d}{dx}\left(4x^2\right) = 8x$$

- **Chain rule:** differentiate each piece separately, then multiply

$$\frac{df}{dx} = \frac{df}{dg} \cdot \frac{dg}{dx} = (2g) \cdot (2) = 4g = 8x$$

```python
x = 3.0
g_val = g(x)

df_dg = 2 * g_val     # derivative of f w.r.t. g, evaluated at g_val
dg_dx = 2.0          # derivative of g w.r.t. x

print(df_dg * dg_dx)
print(8 * x)  # direct derivative, for comparison
```

```
24.0
24.0
```

- Same answer, built from two *simple* pieces instead of one messier one
- This is *the* idea autograd is built on: break a big function into small pieces, differentiate each piece, multiply them together

Partial derivatives: one variable at a time

- Our function now takes more than one input:

$$f(x_1, x_2) = x_1^2 + 2x_2^2$$

- Notation: $\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}$

A **partial derivative** $\frac{\partial f}{\partial x_2}$ asks:

> "How does f change if I move *just* x_2, holding x_1 fixed?"

- $\frac{\partial f}{\partial x_1} = 2x_1$ — treat x_2 as a constant, differentiate like normal
- $\frac{\partial f}{\partial x_2} = 4x_2$ — same move, other variable

The gradient: a vector of partial derivatives

- Given the function:

$$f(x_1, x_2) = x_1^2 + 2x_2^2$$

- Stack the partials into a vector:

$$\nabla f = \left[\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}\right] = [2x_1, 4x_2]$$

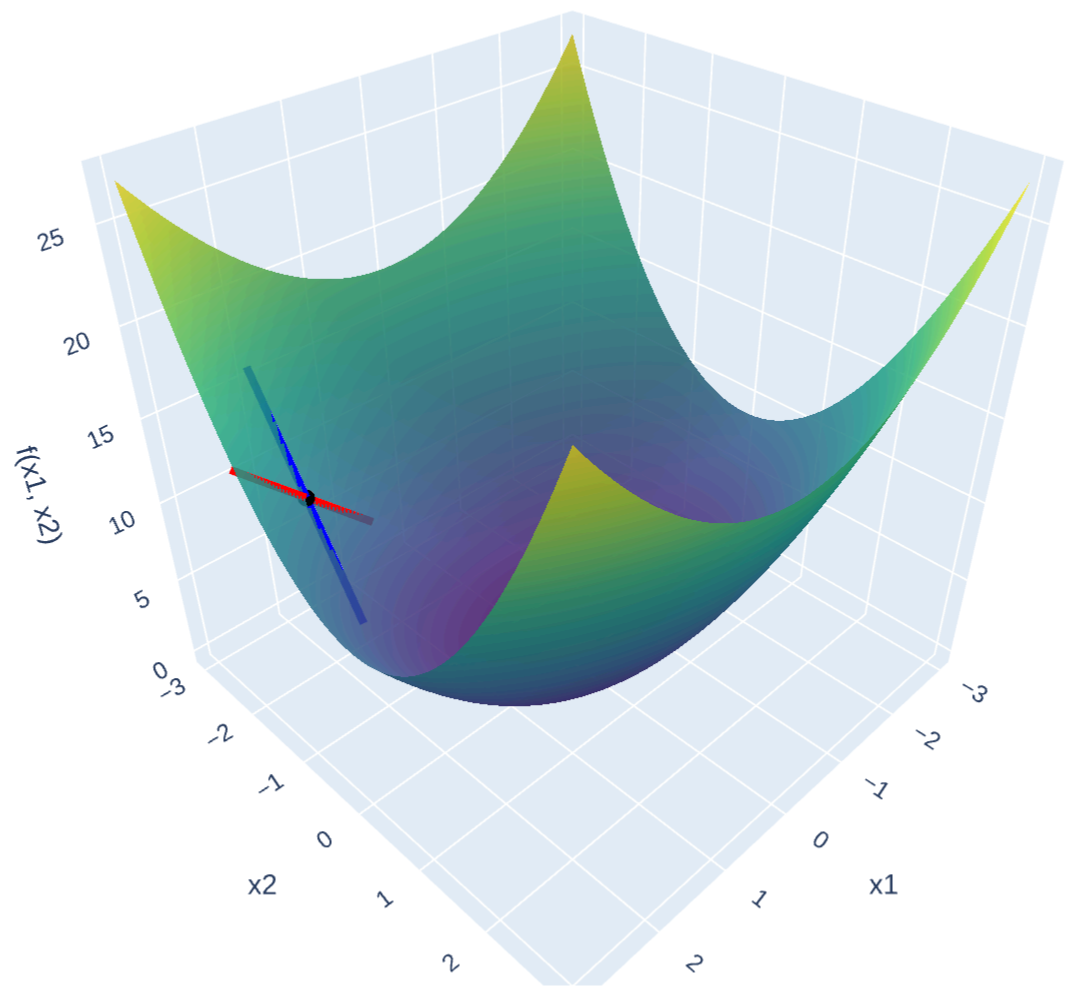
```
def grad_f(x1, x2):
    return torch.tensor([2 * x1, 4 * x2])

print(grad_f(1.0, 1.5))
```

```
tensor([2., 6.])
```

- This vector points toward steepest *increase* — to go downhill, step in the *opposite* direction, $-\nabla f$
- Remember `choose_next_value()`? This is exactly the “smart” direction it was missing

What partial derivatives look like



- **Red line:** the slope along the x_1 direction, holding x_2 fixed — this is $\frac{\partial f}{\partial x_1}$
- **Blue line:** the slope along the x_2 direction, holding x_1 fixed — this is $\frac{\partial f}{\partial x_2}$
- The gradient just bundles these two slopes into a single vector
- Drag to rotate — get a feel for the bowl before we build on it

Try it yourself

Same plot, straight from the slide — paste this into your notebook to rotate it live, move the point, or change the function entirely:

```
import numpy as np
import plotly.graph_objects as go

x1 = np.linspace(-3, 3, 60)
x2 = np.linspace(-3, 3, 60)
X1, X2 = np.meshgrid(x1, x2)
F = X1 ** 2 + 2 * X2 ** 2

p1, p2 = 2.2, -1.8          # try moving this point around
fp = p1 ** 2 + 2 * p2 ** 2
t = np.linspace(-0.8, 0.8, 2)

d1 = 2 * p1  # df/dx1 at (p1, p2)
d2 = 4 * p2  # df/dx2 at (p1, p2)

fig = go.Figure()
fig.add_trace(go.Surface(x=x1, y=x2, z=F, colorscale="Viridis", opacity=0.85, showscale=False))
```